

(PROJECT PROPSAL)



Dust Collector Mini Robot Model

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BACHELOR OF ENGINEERING

IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

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A SENIOR PROJECT SUBMITTED TO
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BACHELOR OF ENGINEERING
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Abstract

This senior project proposes the development of a Bluetooth-enabled home cleaning device, specifically using the Arduino UNO R3 , Motor l298n and Ultrasonic Module HC-SR04 to create a motion-controlled vehicle with an Android smartphone using the created control app. Using MIT INENTOR and downloading it on your Android smartphone, code the robot's motion and distance sensors in Arduino IDE. Connect via Bluetooth HC-05, which will make it more convenient and easier. It can help both experienced and non-tech-savvy users. Have a good understanding and access to the Internet of Things (IoT).

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CHAPTER 1

INTRODUCTION

1.1 Background and Rational

Nowadays, technology plays a more important role in our lives than ever before, whether it is entertainment, education, research. As well as technology is common in today's society known as the Internet of Things (IoT).

Internet of Things (IoT) Internet of Things By various devices, things will be connected to the internet world. Allows humans to control the use of various devices. Through the internet network, such as turning on and off electrical appliances. (Turning on electricity in the home by connecting a control device such as a mobile phone via the internet), cars, mobile phone communications, tools, agricultural tools, buildings, houses, electrical appliances Through the internet network .IoT and with the widespread development of technology. In this study We offer MIT App Inventor that allows users to control simulations via Bluetooth, and this tutorial teaches users how to connect their devices and better access assistive technology. In their daily life.

1.2 Objective

- 1.2.1 Develop steering control system through an app developed by MIT INVENTOR and downloaded to your phone.
- 1.2.2 Create a model that is convenient for the work area
- 1.2.3 To learn the development of a system to help control the robot tell the distance from obstacles.

1.3 Scope

- 1.3.1 The robot can move left, right, forward, backward and tell the distance to obstacles according to the code written in Arduino.
- 1.3.2 when code is written, the CONTROL ROBOT app can connect to the HC-05 Bluetooth.
- 1.3.3 There is an on-off switch, for opening the suction fan

[illegible]

1.6 Expected Result

1.6.1 Control movement on MIT App Inventor with a smartphone and display it on the serial monitor.

1.6.2 Users can learn and understand basic Bluetooth connections by themselves.

1.7 Place and Equipment

1.7.1 Place at Mae Fah Luang University

1.7.2 Hardware

1) arduino uno r3 1Item



2) motor 1298n 2Item



3) battery rail 2Item



4) 4 inch craft vacuum cleaner impeller 1Item



5) DC Motor 3Item



6) dust bag 1Item



7) On/off switch, 2-pin LED light 1Item



8) 2 wheels



9) HC-05 Bluetooth 1Item



10) Ultrasonic Module HC-SR04 2Item



1.7.3 Software

1) arduino IDE



2) App Inventor



CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory

The theory involved for this project is the Internet of Things (IoT), with devices connected to WiFi to control the model via a Bluetooth network. By force on Android mobile to achieve the goal.

The CONTROL ROBOT app developed from the MITInventor app can control the direction of movement: left, right, straight, forward, backward and tells the distance from obstacles in front and behind. by interface design and write code using blocks and upload it to your Android smartphone.

The arduino IDE is a programming tool that is compatible with all versions of the Arduino. We will use the hardware is arduino Uno R3 and write the control code to control the car into the arduino IDE with a Bluetooth HC-05 connected to the robot. The vacuum fan has a separate on-off switch.

2.2 Related Research

Research related to this project can focus on:

Motor Control: Research can be conducted to explore motor control technologies, integrating various electronic devices. To optimize pulse width modulation (PWM) in robot motion control.

Programming: Be able to conduct research to explore different programming languages and frameworks. that can be used to program the robot to control

Strategies for smartphone-controlled robots from the arduino program ide Write the most suitable program for this robot.

Design and Construction: Be able to conduct research to explore design and construction.

Methods for robot components Check out the materials and

Technology that can make robots stable, durable and suitable.

2.3 Related Work

Relevant work for this project may include reviews of existing research and projects related to robotics, including academic papers, articles and online.

Topic-related resources

Examples of relevant work for this project include:

study of "Arduino UNO + L298P Bluetooth Controlled Robot Project" by Ken RobotSiam presents an Android smartphone controlled robot via Bluetooth HC-06, using the L298P Motor Shield.

2.4 Related Technology

Software

C++

C++ is a new and widely used computer programming language. developed from the C language It is a general purpose programming language that supports many programming paradigms. The C language is an intermediate level suitable for structured programming. It is a very flexible language that is compatible with different machines, such as procedural programming, abstraction. object oriented programming general programming and pattern design. This language is faster as compared to other languages and is easy to use. It is commonly used in operating systems, games, business GUIs, web browsers, embedded systems, applications as well as the Arduino IDE

Arduino IDE

Arduino IDE is a tool used to develop programs on the Arduino platform and upload successfully developed programs onto the board. It supports development in many models of boards and can add other boards that are not Arduino boards to be able to develop programs and upload programs onto the board. The Arduino IDE contains various commands. program example and example libraries that has been installed and ready to use More libraries can be downloaded and installed. There is an online cloud for data storage. data can be retrieved and displayed.

App Inventor

is a tool for creating applications for Android smartphones and tablets. It uses the same principle as Scratch but is more complex. The nature of Visual Programming is to write a program by connecting block commands. Focus on designing to solve problems (problem solving) by creating programs on mobile phones, smart phones. .Users develop Android mobile applications by creating projects and programming on web browsers. that connects to App Inventor servers when programmed You can test it with a mobile emulator (Android emulator) or a real Android mobile phone.

Hardware

Arduino UNO R3

Arduino UNO R3 is an AVR microcontroller board that has been developed by Open Source, with both hardware and software information disclosed. The Arduino board is designed to be easy to use. The ATmega328P chip-based microcontroller board features 14 digital input/output pins, 6 analog inputs, a 16 MHz quartz crystal, a USB connection, and a power jack. The board is widely used in many DIY projects, including robotics.

HC-05 Bluetooth

It is a Bluetooth module that is used to connect to various smart devices. Allows smart devices to communicate with microcontrollers (Arduino AVR PIC etc.) via Serial port. HC05 module can be set to use in both Master mode (allowing other devices to connect) and Mode. Slave (connected to other devices) Settings such as device name, password can be done via AT Command, which requires a special pin to be connected in order for the module to enter the setting mode. or press and hold the button on the module

L298N

is a motor control module. It is an H-bridge type driver that allows controlling the speed and direction of rotation of a DC motor. It can also be easily used with stepper motors.

Ultrasonic sensor

It is a sensor that uses the principle of reflecting ultrasonic frequencies to detect various objects. Ultrasonic sensors require a moving medium such as air, gas, or liquid. Ultrasonic sensors can be used to detect many types of objects. Objects that are in a liquid state, such as chemicals or viscosities. Ultrasonic sensors can be used for detection. And ultrasonic sensors have a frequency of 20,000Hz or more, which is a higher frequency than humans can perceive.

CHAPTER 3

ANALYSIS AND DESIGN

3.1 FlowChart

A flowchart created with the vision that users need to start the CONTROL ROBOT application, an application we designed and built. To control this robot, design it in the IMT INVENTOR app and code it in the ARDUINO IDE, then open the phone application. Bluetooth connection can move forward, backward, left, right and tell the distance to obstacles. As for the on-off function of the exhaust fan, we have a switch attached to the robot for convenience in controlling the direction and according to the user's target distance.

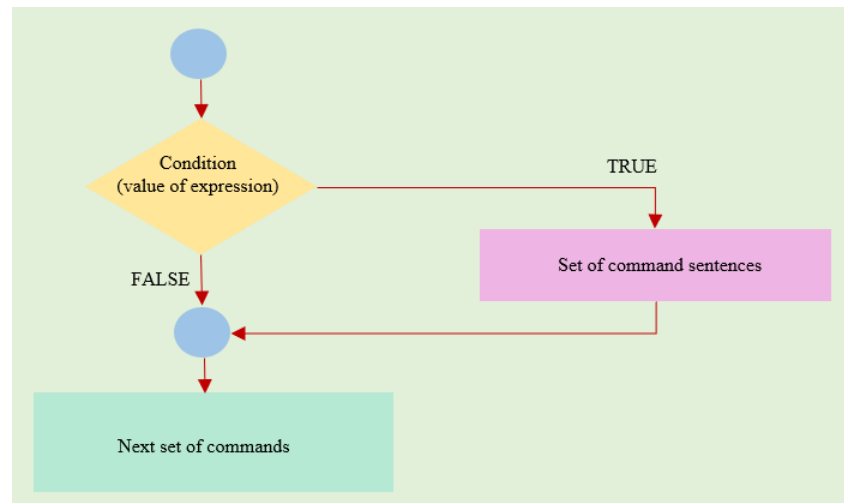


Figure 3.1 The system operation



Figure 3.1 The Interfaces and code blocks in MIT inventor apps

3.2 Electronic Configurations

In addition to the invention designed to control the robot. One of our project is signal transmission and electronic connectivity.

Code 1 :Open the Arduino program (IDE), write code and upload it to the Arduino UNO board for HC-05 Bluetooth. Connect to bluetooth as follows.

```
#include <SoftwareSerial.h>

SoftwareSerial BTSerial(9, 10);

void setup()
{
  pinMode(8, OUTPUT);
  digitalWrite(8, HIGH);
  Serial.begin(9600);
  BTSerial.begin(9600);
  delay(1000);
  Serial.println("Enter AT commands:");
}

void loop()
{
  if (BTSerial.available())
    Serial.write(BTSerial.read());

  if (Serial.available())
    BTSerial.write(Serial.read());
}
```

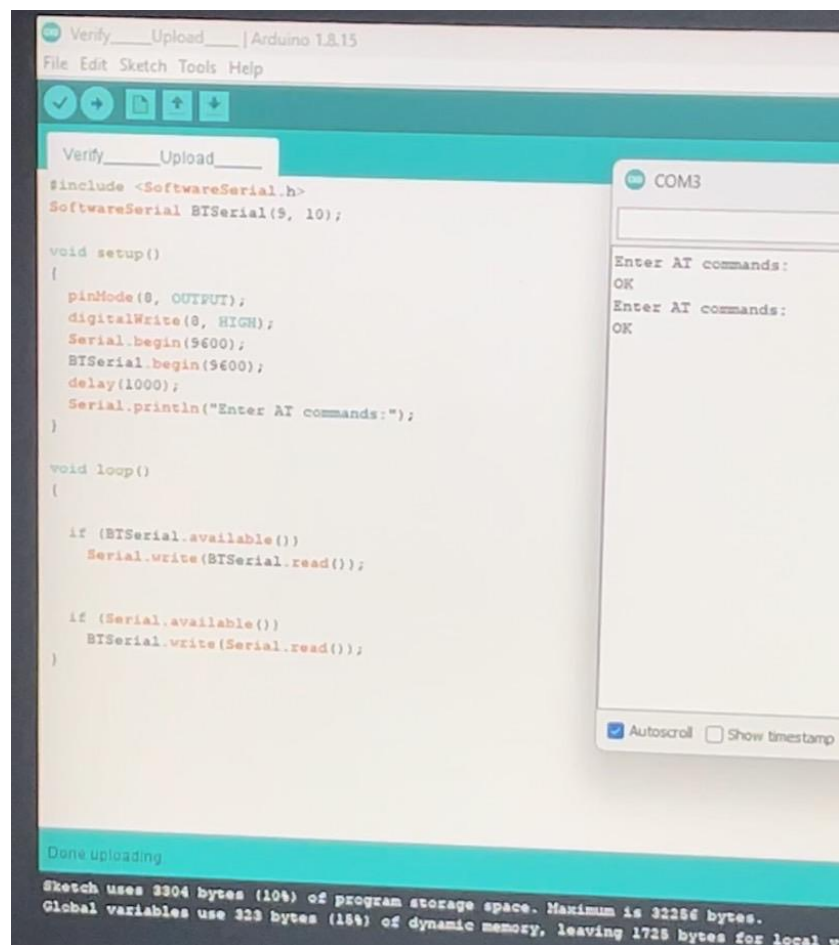


Figure 3.2 Code for HC-05 Bluetooth. Connect to bluetooth

Code 2 :Open the Arduino program (IDE), write code for left, right, front, back, and add 2 distance sensors and upload them to the Arduino UNO board as follows.

```
#include <SoftwareSerial.h>

SoftwareSerial BTSerial(9, 10);

int dir1PinA = 2;
int dir2PinA = 3;
int speedPinA = 6;
int dir1PinB = 4;
int dir2PinB = 5;
int speedPinB = 7;

// Define trigger and echo pins for the two ultrasonic sensors
int trigPin1 = 12;
int echoPin1 = 11;
int trigPin2 = 13;
int echoPin2 = 1;

void setup()
{
  Serial.begin(9600);
  pinMode(dir1PinA, OUTPUT);
  pinMode(dir2PinA, OUTPUT);
  pinMode(speedPinA, OUTPUT);
  pinMode(dir1PinB, OUTPUT);
  pinMode(dir2PinB, OUTPUT);
  pinMode(speedPinB, OUTPUT);

  // Set up ultrasonic sensor pins
```

```
pinMode(trigPin1, OUTPUT);
pinMode(echoPin1, INPUT);
pinMode(trigPin2, OUTPUT);
pinMode(echoPin2, INPUT);

pinMode(8, OUTPUT);
digitalWrite(8, HIGH);
Serial.begin(9600);
BTSerial.begin(9600);
}

void loop()
{
  if (BTSerial.available())
    Serial.write(BTSerial.read());

  if (Serial.available())
    BTSerial.write(Serial.read());

  if (BTSerial.available() > 0)
  {

    int inByte = BTSerial.read();
    int speed;
    switch (inByte)
    {

    case 'F':
      analogWrite(speedPinA, 255);
      analogWrite(speedPinB, 255);
      digitalWrite(dir1PinA, LOW);
```

```
digitalWrite(dir1PinB, HIGH);  
digitalWrite(dir2PinA, HIGH);  
digitalWrite(dir2PinB, LOW);  
Serial.println("Motor 1 Forward");  
Serial.println("Motor 2 Forward");  
Serial.println(" ");  
break;
```

case 'S':

```
analogWrite(speedPinA, 0);  
digitalWrite(dir1PinA, LOW);  
digitalWrite(dir2PinA, HIGH);  
Serial.println("Motor 1 Stop");  
analogWrite(speedPinB, 0);  
digitalWrite(dir1PinB, LOW);  
digitalWrite(dir2PinB, HIGH);  
Serial.println("Motor 2 Stop");  
Serial.println(" ");  
break;
```

case 'B':

```
analogWrite(speedPinA, 255);  
digitalWrite(dir1PinA, HIGH);  
digitalWrite(dir2PinA, LOW);  
Serial.println("Motor 1 Back");  
analogWrite(speedPinB, 255);  
digitalWrite(dir1PinB, LOW);  
digitalWrite(dir2PinB, HIGH);  
Serial.println("Motor 2 Back");  
Serial.println(" ");
```

```
break;
```

```
case 'L':
```

```
    analogWrite(speedPinA, 0);  
    digitalWrite(dir1PinA, LOW);  
    digitalWrite(dir2PinA, HIGH);  
    Serial.println("Motor 1 Left");  
    analogWrite(speedPinB, 255);  
    digitalWrite(dir1PinB, HIGH);  
    digitalWrite(dir2PinB, LOW);  
    Serial.println("Motor 2 Left");  
    Serial.println(" ");  
    break;
```

```
case 'R':
```

```
    analogWrite(speedPinA, 255);  
    digitalWrite(dir1PinA, LOW);  
    digitalWrite(dir2PinA, HIGH);  
    Serial.println("Motor 1 Right");  
    analogWrite(speedPinB, 0);  
    digitalWrite(dir1PinB, LOW);  
    digitalWrite(dir2PinB, HIGH);  
    Serial.println("Motor 2 Right");  
    Serial.println(" ");  
    break;
```

```
case 'T':
```

```
    analogWrite(speedPinA, 150);  
    digitalWrite(dir1PinA, LOW);  
    digitalWrite(dir2PinA, HIGH);
```

```
Serial.println("Motor 1 Forward L");  
analogWrite(speedPinB, 255);  
digitalWrite(dir1PinB, HIGH);  
digitalWrite(dir2PinB, LOW);  
Serial.println("Motor 2 Forward L");  
Serial.println(" ");  
break;
```

```
case 'G':  
    analogWrite(speedPinA, 255);  
    digitalWrite(dir1PinA, LOW);  
    digitalWrite(dir2PinA, HIGH);  
    Serial.println("Motor 1 Forward R");  
    analogWrite(speedPinB, 150);  
    digitalWrite(dir1PinB, HIGH);  
    digitalWrite(dir2PinB, LOW);  
    Serial.println("Motor 2 Forward R");  
    Serial.println(" ");  
    break;
```

```
case 'J':  
    analogWrite(speedPinA, 200);  
    digitalWrite(dir1PinA, HIGH);  
    digitalWrite(dir2PinA, LOW);  
    Serial.println("Motor 1 Back L");  
    analogWrite(speedPinB, 255);  
    digitalWrite(dir1PinB, LOW);  
    digitalWrite(dir2PinB, HIGH);  
    Serial.println("Motor 2 Back L");  
    Serial.println(" ");
```

```

    break;

case 'H':
    analogWrite(speedPinA, 255);
    digitalWrite(dir1PinA, HIGH);
    digitalWrite(dir2PinA, LOW);
    Serial.println("Motor 1 Back R");
    analogWrite(speedPinB, 200);
    digitalWrite(dir1PinB, LOW);
    digitalWrite(dir2PinB, HIGH);
    Serial.println("Motor 2 Back R");
    Serial.println(" ");
    break;

default:
    for (int thisPin = 2; thisPin < 11; thisPin++)
    {
        digitalWrite(thisPin, LOW);
    }
}

// Read distance from the first sensor
long duration1, distance1;
digitalWrite(trigPin1, LOW);
delayMicroseconds(2);
digitalWrite(trigPin1, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin1, LOW);
duration1 = pulseIn(echoPin1, HIGH);
distance1 = (duration1 / 2) / 29.1; // Calculate distance in cm

```

```

// Read distance from the second sensor
long duration2, distance2;

digitalWrite(trigPin2, LOW);

delayMicroseconds(2);

digitalWrite(trigPin2, HIGH);

delayMicroseconds(10);

digitalWrite(trigPin2, LOW);

duration2 = pulseIn(echoPin2, HIGH);

distance2 = (duration2 / 2) / 29.1; // Calculate distance in cm

// Print distance readings

Serial.print("Distance Sensor back: ");

Serial.print(distance1);

Serial.print(" cm\t");

Serial.print("Distance Sensor front: ");

Serial.print(distance2);

Serial.println(" cm");

// Add a delay of 100 milliseconds

delay(100);

}

```

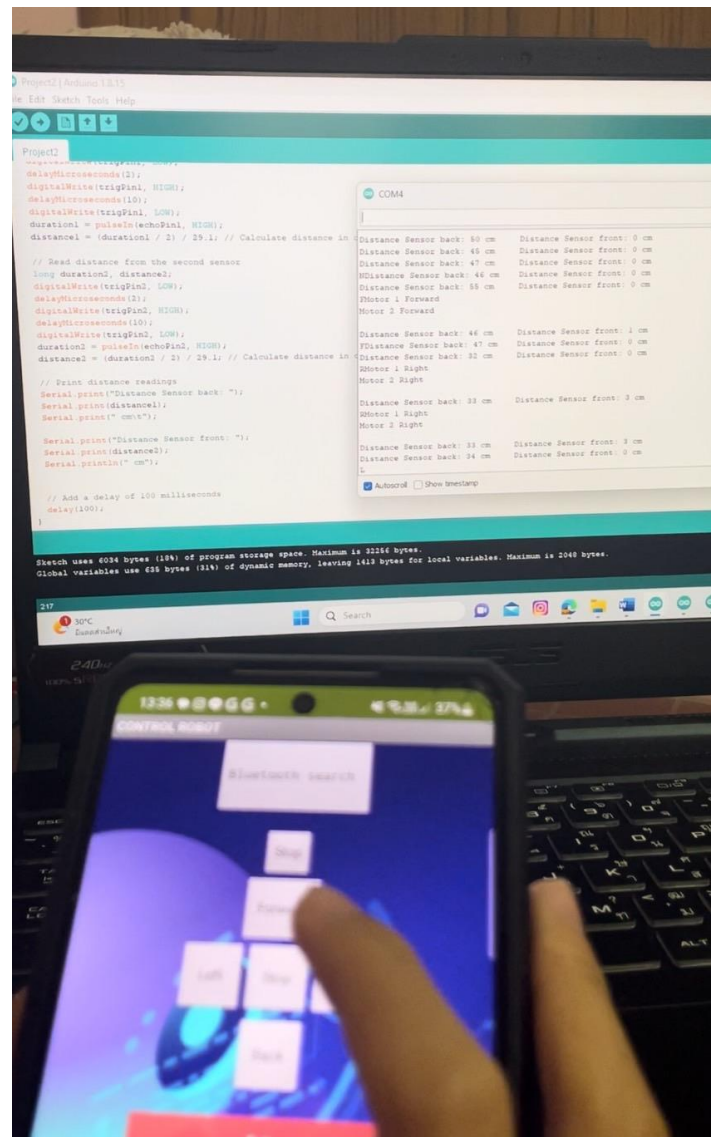


Figure 3.2 Code: left, right, front, back, and distance

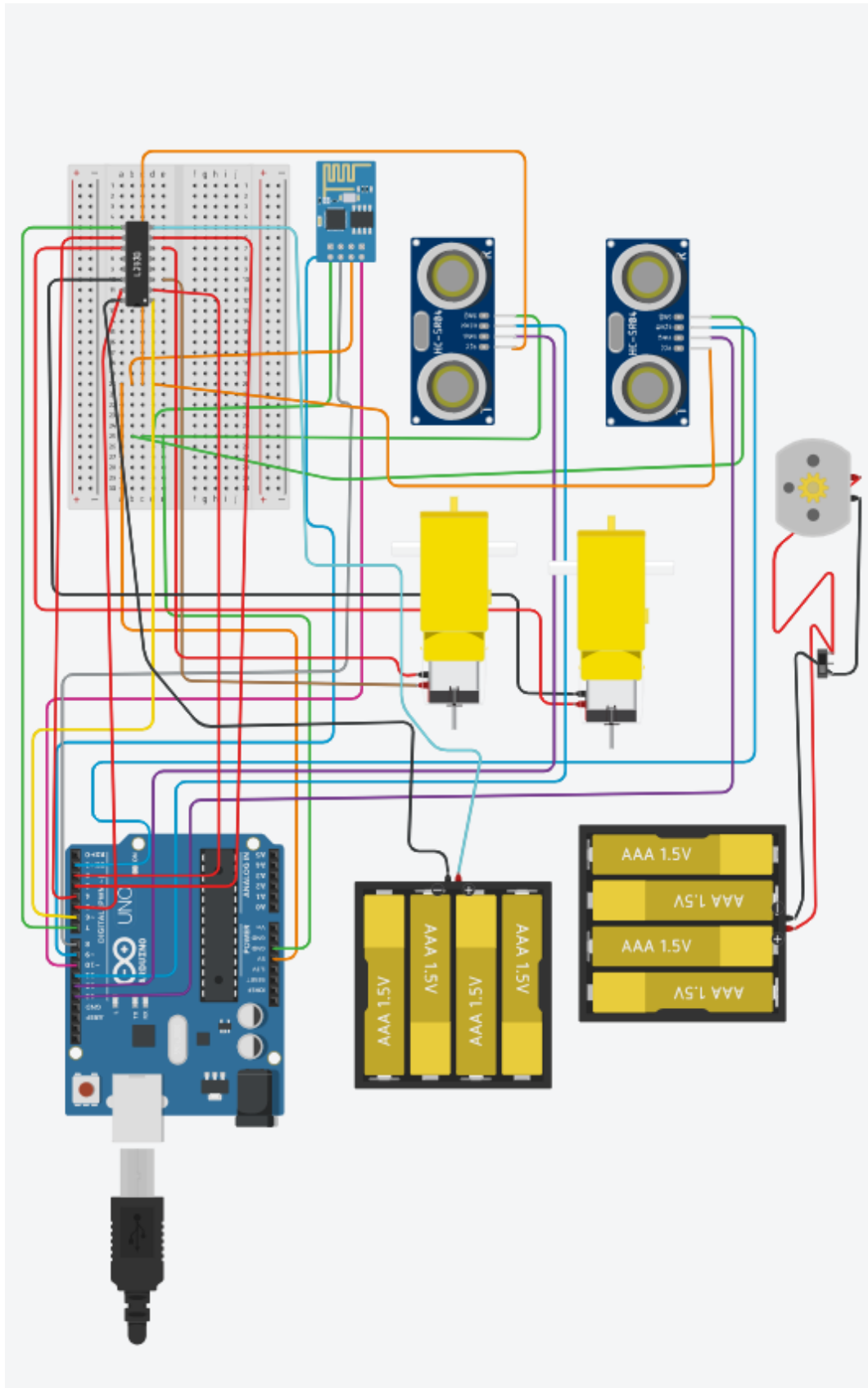


Figure 3.2 Circuit Design

3.3 Schematic diagram of Dust Collector Mini Robot

This Dust Collector Mini Robot has a total of 3 parts, counting from bottom to top. The first part at the bottom is the wheel and propeller for vacuuming dust, the second part is the circuit board layer, and the third part is the dust box like this model.

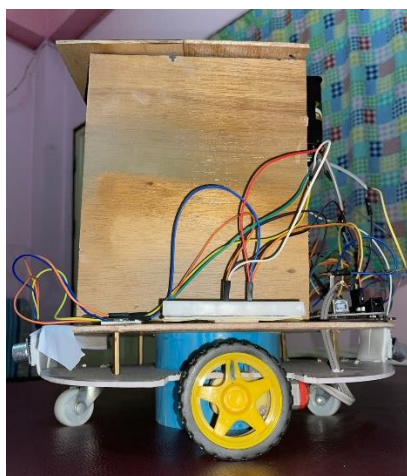
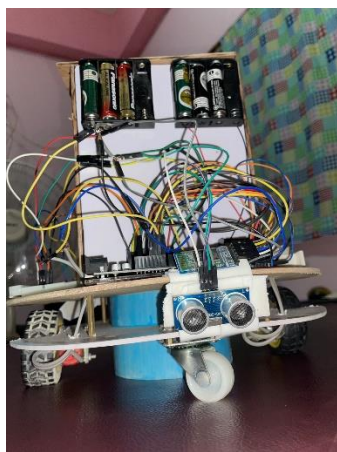
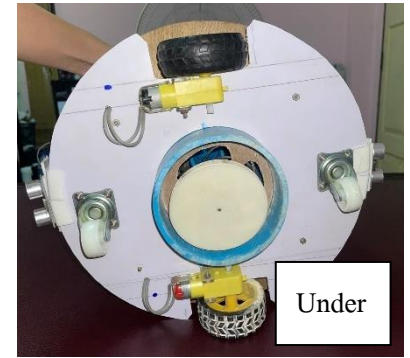
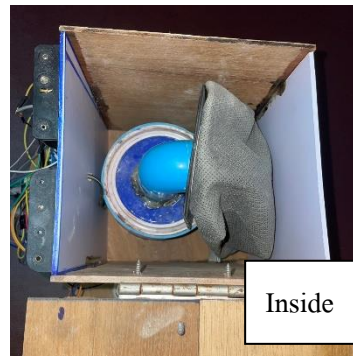


Figure 3.3 Dust Collector Mini Robot Model design

CHAPTER 4

CONCLUSION

4.1 Conclusion

This Dust Collector Mini Robot Model and The CONTROL ROBOT application via Bluetooth are tools to help lighten cleaning. This device has a function that facilitates forced cleaning via the phone. Can control this robot by yourself without having to waste effort to clean. It also allows users to enjoy cleaning.

4.2 Discussion

Currently, this project is still in the research and study stage. because the movement is not yet complete Structural equipment is overweight, however, the project is proceeding smoothly. and waiting for field testing and further project implementation Currently the project is having a few issues while forcing the model to move according to the mobile phone. To solve this problem We will change the material that makes this Dust Collector Mini Robot Model.

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